

Antonio Badilla Olivas

AI/ML Engineer, Independent Researcher, figuring out how to make AI go well.

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SUMMARY

I build, evaluate and monitor AI systems in production, lead AI enablement for engineering teams, and run independent research on how language models behave: evaluations, red-teaming, interpretability and model forensics.

EXPERIENCE

SPAR Research Fellow, In-the-Wild AI Control

Sep 2026 – Dec 2026

Supervised Program for Alignment Research

Remote

- Twelve-week research project with Sree Sharvesh and Thao Pham (MATS) on red-teaming environments where attacks adapt to the monitors that watch them.

Biosecurity Fellow

Aug 2026 – Sep 2026

BlueDot Impact

Remote, United Kingdom

CAMBRIA Fellow

Jul 2026

Cambridge Boston Alignment Initiative

New York, United States

- Selected for a three-week ML research bootcamp built on the ARENA curriculum. The capstone, with two collaborators, replicated Anthropic's emotion vectors on Gemma 4 31B and tested them as a monitor and as a control. Per-token read-out never reached 50% accuracy, against a chance line of one in twelve, and steering moved behaviour to 1–3% of the published effect. The steering null was pre-registered and held across a 4x dose escalation.
- Bulk generation for the capstone ran 17x faster on offline vLLM, which I brought up on an unsupported Blackwell GPU by switching attention backends. Profiling the activation-extraction path cut model load from 33 minutes to 46 seconds.

Lead AI Engineer (Contract)

May 2025 – Present

Colibri Group

Remote, United States

- Built the controls and monitoring for the company's production agents, which run under live user traffic.
- Built automated red-teaming of the deployed agents on Anthropic's Bloom. These audit runs search for concerning behaviour, such as sycophancy and harmfulness, and characterise what they find.
- Led the offline evaluation infrastructure that catches regressions across the company's AI systems, and received an AWS Champions Award in 2026 for the architecture behind Agent Hub, a platform where non-technical users build, evaluate and deploy agents through conversation, running on AWS Agent Core.
- Designed, built and evaluated the company's agents: conversational agents for purchase advice, tutoring and coaching, long-running agents for systematic literature reviews over National Library of Medicine archives, and a semantic search service over unstructured knowledge bases behind one API. The open-weights models behind them are served with vLLM.

Lead AI/ML Engineer

May 2025 – Present

Costa Rica Software Services (CRSS)

Remote, Costa Rica

- Built the core classifier behind tracepipe.ai, which turns long, messy logs of real users' browser sessions into narratives used to adapt agents through examples, skills and fine-tuning.
- Design and run model capability evaluations to establish feasibility before anything is built, and lead the teams that architect, build and evaluate AI systems for internal use and external clients.
- Lead AI enablement for the engineering team, and have trained 50+ engineers in agentic software engineering and published a five-session course on it. Work daily in Claude Code and Codex with my own hooks, skills and subagents.

AGI Strategy & Technical AI Safety Fellow

Feb 2026 – May 2026

BlueDot Impact

Remote, United Kingdom

- Surveyed evaluations and red-teaming, AI control, scalable oversight and interpretability. Wrote step-by-step "kill chain" scenarios for how advanced AI could cause harm and where the safeguards sit across the deployment stack.

Professor (Part-time)

Aug 2025 – Jan 2026

School of Computer Science and Informatics, UCR

San José, Costa Rica

- Taught Python programming to undergraduate students.

AI & Automation Engineer

Mar 2025 – May 2025

Moody's

Heredia, Costa Rica

- Built agents for data processing and automation, and worked on AI enablement for the team.

Lecturer in AI and ML

Dec 2024 – Apr 2025

School of Computer Science and Informatics, UCR

San José, Costa Rica

- Delivered six hands-on workshops on AI engineering and ML mathematics to 200+ students, with practical coursework in LangGraph, PyTorch and scikit-learn.

Data Engineer

Aug 2024 – Feb 2025

Graduate Studies System (SEP), UCR

San José, Costa Rica

- Gathered data-extraction and warehousing requirements from the graduate-affairs system's stakeholders, then migrated the legacy database to a normalised PostgreSQL schema and built the data pipelines.

Teaching Assistant

School of Computer Science and Informatics, UCR

Mar 2022 – Dec 2024

San José, Costa Rica

- Assisted in five undergraduate courses, from Introduction to Computer Science to Artificial Intelligence.

Research Assistant

Research Center in Information and Communication Technologies (CITIC), UCR

Mar 2021 – May 2022

San José, Costa Rica

- Researched mobile development for rehabilitation and built a Flutter app to accessibility guidelines (conference paper).

RESEARCH AND PROJECTS

Odd Number: model forensics of in-context reward hacking | *Python. Trace explorer*

Aug – Sep 2026

- A study of why reasoning models answer odd when the user asks for an even number and an in-context grader pays for odd. Across nine open-weight models at 40 rollouts each, two game the grader; Qwen 3.8 27B does so in 6 of 40 runs and in 0 of 40 when the grader agrees with the instruction. Reading 360 of its chains of thought, with a verbatim quote required for every label, the question that predicts an odd answer is whether the sender is a person or a machine. One added sentence saying the user wrote the instruction takes the rate to 0 of 40. Every trace is readable in a browser explorer.

daemon: a monitor for long-running Claude Code sessions | *TypeScript, Claude Agent SDK. Open source*

Mar 2026

- A proof-of-concept monitor for Claude Code sessions that run for hours. It collects every hook event and OpenTelemetry log into SQLite, then an Agent SDK analysis reads the whole session and reports each failure with its type, its impact and the event IDs that prove it, ignoring retries that succeeded. Recurring failures become recommendations for changing the harness. Daemon watched the agents that built it.

Research Engineering Harness | *DOI 10.5281/zenodo.21617975*

2026

- A portable harness for research done with coding agents. It ships nine skills for Claude Code and Codex. Its checks fail the build when a claim cites evidence that does not exist, when a number in a write-up disagrees with its results file, or when a claim is marked *survived* without a falsification attempt.

Secret Hitler Sandbox | *MCP, FastAPI. DOI 10.5281/zenodo.21618869*

Mar 2026 – Present

- A sandbox where LLM agents play a full game of Secret Hitler against each other over tool calls, a game you win by deceiving the other players. A deterministic engine enforces the rules and logs every action and private message, so who deceived whom can be reconstructed afterwards.

The Refusal Axis: Domain Decomposition in Gemma 3 12B | *GemmaScope 2 SAEs. DOI 10.5281/zenodo.21617969*

2026

- A white-box study of whether the single refusal direction described by Arditi et al. (2024) hides several domain-specific directions. Refusal occupies a structured 11-dimensional subspace in the residual stream ($p < 0.001$ against random-vector controls). Capping along the safety direction at layer 36 cut safety refusal by 31.6 points on a 0–100 scale while capability, privacy and benign responses moved within ± 1.3 .

Prompt Format and Guardrail Compliance in Small LLMs | *Gemma 3, SAEs. Open source*

Aug 2025 – Mar 2026

- A study of whether the format of a system prompt changes how well a small agent honours its guardrails. It ran 2,400 paired runs of Gemma 3 4B and 12B RAG agents, scored by LLM judges for refusal, leakage and groundedness. The main finding is that refusing and not leaking are different skills: among correct refusals, 74.8% of the 4B model's answers still leak private information, against 48.2% for the 12B.

Agentic Software Engineering Course | *published at antonio-tresol.github.io*

2026

- A five-session course on agentic software engineering and AI enablement: getting the most autonomous work out of coding agents while holding the bar on quality, security and taste, through context engineering, MCP and multi-agent coordination.

EDUCATION

Master of Science in Computer Science and Informatics (paused)

Universidad de Costa Rica

Mar 2025 – Jul 2026

San Pedro, San José

- Coursework and early thesis work in mechanistic interpretability and AI alignment (transformer internals, sparse autoencoders).

Bachelor of Science in Computer Science (GPA: 9.17 / 10)

Universidad de Costa Rica

Mar 2021 – Dec 2024

San Pedro, San José

- Coursework: Machine Learning, Artificial Intelligence, Probability and Statistics, Experiment Design. Conference paper on an ensemble ViT model for Costa Rican flora that beat the previous best published result by 5%, *Revista Ingeniería (UCR)*.

TECHNICAL SKILLS

Programming: Python, TypeScript, SQL, Java, C/C++, JavaScript.

ML and inference: PyTorch, HuggingFace Transformers, scikit-learn, W&B, fine-tuning, SAEs, steering, vLLM (serving, batched generation).

Evaluation and monitoring: LLM judges, eval harnesses, regression testing, red-teaming (Bloom), production monitoring, Inspect.

Agents: Claude Code, Codex, OpenCode, Claude Agent SDK, MCP, LangChain, LangGraph, multi-agent environments.

Infrastructure: Git, Docker, AWS, Azure, Linux, FastAPI, PostgreSQL, SQLite, GitHub Actions CI/CD.

Spoken languages: English C1 (IELTS certified), French C1, Spanish native.